2.1 Plan: Refer to the definitions of an element and a compound.
Solution:
Unlike compounds, elements cannot be broken down by chemical changes into simpler materials. Compounds contain different types of atoms; there is only one type of atom in an element.

2.4 Plan: Review the definitions of elements, compounds and mixtures.
Solution:
a) The presence of more than one element (calcium and chlorine) makes this pure substance a **compound**.
b) There are only atoms from one element, sulfur, so this pure substance in an **element**.
c) The presence of more than one compound makes this a **mixture**.
d) The presence of more than one type of atom means it cannot be an element. The specific, not variable, arrangement means it is a **compound**.

2.12 Plan: Restate the three laws in your own words.
Solution:
a) The law of mass conservation applies to all substances — **elements, compounds, and mixtures**. Matter can neither be created nor destroyed, whether it is an element, compound, or mixture.
b) The law of definite composition applies to **compounds** only, because it refers to a constant, or definite, composition of elements within a compound.
c) The law of multiple proportions applies to **compounds** only, because it refers to the combination of elements to form compounds.

2.14 Plan: Review the three laws: Law of Mass Conservation, Law of Definite Composition, and Law of Multiple Proportions.
Solution:
a) Law of Definite Composition — The compound potassium chloride, KCl, is composed of the same elements and same fraction by mass, regardless of its source (Chile or Poland).
b) Law of Mass Conservation — The mass of the substance inside the flashbulb did not change during the chemical reaction (formation of magnesium oxide from magnesium and oxygen).
c) Law of Multiple Proportions — Two elements, O and As, can combine to form two different compounds that have different proportions of As present.

2.18 Plan: Review the mass laws: Law of Mass Conservation, Law of Definite Composition, and Law of Multiple Proportions.
Solution:
Experiments 1 and 2 together demonstrate the **Law of Definite Composition**. When 3.25 times the amount of blue compound in experiment 1 is used in experiment 2, then 3.25 times the amount of products were made and the relative amounts of each product are the same in both experiments. In experiment 1, the ratio of white compound to colorless gas is 0.64:0.36 or 1.78:1 and in experiment 2, the ratio is 2.08:1.17 or 1.78:1. The two experiments also demonstrate the **Law of Conservation of Mass** since the total mass before reaction equals the total mass after reaction.

2.20 Plan: The difference between the mass of fluorite and the mass of calcium gives the mass of fluorine. The masses of calcium, fluorine, and fluorite combine to give the other values.
Solution:
Fluorite is a mineral containing only calcium and fluorine.
a) Mass of fluorine = mass of fluorite – mass of calcium = 2.76 g – 1.42 g = **1.34 g fluorine**

b) To find the mass fraction of each element, divide the mass of each element by the mass of fluorite:

$$\text{Mass fraction of Ca} = \frac{\text{mass Ca}}{\text{mass fluorite}} = \frac{1.42 \text{ g Ca}}{2.76 \text{ g fluorite}} = 0.51449 = \mathbf{0.514}$$

$$\text{Mass fraction of F} = \frac{\text{mass F}}{\text{mass fluorite}} = \frac{1.34 \text{ g F}}{2.76 \text{ g fluorite}} = 0.48551 = \mathbf{0.486}$$

c) To find the mass percent of each element, multiply the mass fraction by 100:
Mass % Ca = (0.514) (100) = 51.449 = **51.4%**
Mass % F = (0.486) (100) = 48.551 = **48.6%**

2.22 Plan: Dividing the mass of magnesium by the mass of the oxide gives the ratio. Use this ratio, and the mass of the second sample to determine the mass of magnesium.
Solution:
a) If 1.25 g of MgO contains 0.754 g of Mg, then the mass ratio (or fraction) of magnesium in the oxide compound is $\frac{\text{mass Mg}}{\text{mass MgO}} = \frac{0.754 \text{ g Mg}}{1.25 \text{ g MgO}} = 0.6032 = \mathbf{0.603}.$

b) Mass of magnesium $= (534 \text{ g MgO})\left(\frac{0.6032 \text{ g Mg}}{1 \text{ g MgO}}\right) = 322.109 = \mathbf{322 \text{ g magnesium}}$

2.24 Plan: Since copper is a metal and sulfur is a nonmetal, the sample contains 88.39 g Cu and 44.61 g S. Calculate the mass fraction of each element in the sample.
Solution:
Mass of compound = 88.39 g copper + 44.61 g sulfur = 133.00 g compound

$$\text{Mass of copper} = (5264 \text{ kg compound})\left(\frac{10^3 \text{ g compound}}{1 \text{ kg compound}}\right)\left(\frac{88.39 \text{ g copper}}{133.00 \text{ g compound}}\right) = 3.49838 \times 10^6$$
$$= \mathbf{3.498 \times 10^6 \text{ g copper}}$$

$$\text{Mass of sulfur} = (5264 \text{ kg compound})\left(\frac{10^3 \text{ g compound}}{1 \text{ kg compound}}\right)\left(\frac{44.61 \text{ g sulfur}}{133.00 \text{ g compound}}\right) = 1.76562 \times 10^6$$
$$= \mathbf{1.766 \times 10^6 \text{ g sulfur}}$$

2.26 Plan: The law of multiple proportions states that if two elements form two different compounds, the relative amounts of the elements in the two compounds form a whole number ratio. To illustrate the law we must calculate the mass of one element to one gram of the other element for each compound and then compare this mass for the two compounds. The law states that the ratio of the two masses should be a small whole number ratio such as 1:2, 3:2, 4:3, etc.

Solution:

Compound 1: $$\frac{47.5 \text{ mass \% S}}{52.5 \text{ mass \% Cl}} = 0.90476 = 0.905$$

Compound 2: $$\frac{31.1 \text{ mass \% S}}{68.9 \text{ mass \% Cl}} = 0.451379 = 0.451$$

Ratio: $$\frac{0.905}{0.451} = 2.0067 = 2.00 / 1.00$$

Thus, the ratio of the mass of sulfur per gram of chlorine in the two compounds is a small whole number ratio of 2 to 1, which agrees with the law of multiple proportions.

2.29 Plan: Determine the mass percent of sulfur in each sample by dividing the grams of sulfur in the sample by the total mass of the sample. The coal type with the smallest mass percent of sulfur has the smallest environmental impact.

Solution:

$$\text{Mass \% in Coal A} = \left(\frac{11.3 \text{ g sulfur}}{378 \text{ g sample}}\right)(100\%) = 2.9894 = \mathbf{2.99\%\ S\ (by\ mass)}$$

$$\text{Mass \% in Coal B} = \left(\frac{19.0 \text{ g sulfur}}{495 \text{ g sample}}\right)(100\%) = 3.8384 = \mathbf{3.84\%\ S\ (by\ mass)}$$

$$\text{Mass \% in Coal C} = \left(\frac{20.6 \text{ g sulfur}}{675 \text{ g sample}}\right)(100\%) = 3.0519 = \mathbf{3.05\%\ S\ (by\ mass)}$$

Coal A has the smallest environmental impact.

2.36 Plan: Re-examine the definitions of atomic number and the mass number.

Solution:

The atomic number is the number of protons in the nucleus of an atom. When the atomic number changes, the identity of the element also changes. The mass number is the total number of protons and neutrons in the nuclues of an atom. Since the identity of an element is based on the number of protons and not the number of neutrons, the mass number can vary (by a change in number of neutrons) without changing the identity of the element.

2.39 Plan: The superscript is the mass number. Consult the periodic table to get the atomic number (the number of protons). The mass number – the number of protons = the number of neutrons. For atoms, the number of protons and electrons are equal.

Solution:

Isotope	Mass Number	# of Protons	# of Neutrons	# of Electrons
^{36}Ar	36	18	18	18
^{38}Ar	38	18	20	18
^{40}Ar	40	18	22	18

2.47 Plan: To calculate the atomic mass of an element, take a weighted average based on the natural abundance of the isotopes: (isotopic mass of isotope 1 x fractional abundance) + (isotopic mass of isotope 2 x fractional abundance).
Solution:

$$\text{Atomic mass of gallium} = (68.9256\text{ amu})\left(\frac{60.11\%}{100\%}\right) + (70.9247\text{ amu})\left(\frac{39.89\%}{100\%}\right) = 69.7230 = \mathbf{69.72\ amu}$$

2.49 Plan: To find the percent abundance of each Cl isotope, let x equal the fractional abundance of ^{35}Cl and $(1 - x)$ equal the fractional abundance of ^{37}Cl. Remember that atomic mass =(isotopic mass of ^{35}Cl x fractional abundance) + (isotopic mass of ^{37}Cl x fractional abundance)
Solution:
Atomic mass of Cl = 35.4527 amu

$$35.4527 = 34.9689x + 36.9659(1 - x)$$
$$35.4527 = 34.9689x + 36.9659 - 36.9659x$$
$$35.4527 = 36.9659 - 1.9970x$$
$$1.9970x = 1.5132$$
$$x = 0.75774 \text{ and } 1 - x = 0.24226$$

% abundance ^{35}**Cl = 75.774%** % abundance ^{37}**Cl = 24.226%**

2.58 Plan: Review the section in the chapter on the periodic table. Remember that alkaline earth metals are in Group 2A(2) and the halogens are in Group 7A(17); periods are horizontal rows.
Solution:
a) The symbol and atomic number of the heaviest alkaline earth metal are **Ra** and **88**.
b) The symbol and atomic number of the lightest metalloid in Group 4A(14) are **Si** and **14**.
c) The symbol and atomic mass of the coinage metal whose atoms have the fewest electrons are **Cu** and **63.55 amu**.
d) The symbol and atomic mass of the halogen in Period 4 are **Br** and **79.90 amu**.

2.60 Plan: Review the section of the chapter on the formation of ionic compounds.
Solution:
These atoms will form **ionic** bonds, in which one or more electrons are transferred from the metal atom to the nonmetal atom to form a cation and an anion, respectively. The oppositely charged ions attract, forming the ionic bond.

2.74 Plan: Determine the charges of the ions based on their position on the periodic table. Next, determine the ratio of the charges to get the ratio of the ions.
Solution:
Lithium [Group 1A(1)] forms the Li^+ ion; oxygen [Group 6A(16)] forms the O^{2-} ion $(6 - 8 = -2)$. The ionic compound that forms from the combination of these two ions must be electrically neutral, so two Li^+ ions combine with one O^{2-} ion to form the compound Li_2O. There are twice as many Li^+ ions as O^{2-} ions in a sample of Li_2O.

$$\text{Number of } O^{2-} \text{ ions} = (8.4 \times 10^{21}\ Li^+\text{ ions})\left(\frac{1\ O^{2-}\text{ ion}}{2\ Li^+\text{ ions}}\right) = \mathbf{4.2 \times 10^{21}\ O^{2-}\ ions}$$

2.76 Plan: The key is the size of the two alkali ions. The charges on the sodium and potassium ions are the same as both are in Group 1A(1), so there will be no difference due to the charge. The chloride ions are the same, so there will be no difference due to the chloride.
Solution:
Coulomb's law states that the energy of attraction in an ionic bond is directly proportional to the *product of charges* and inversely proportional to the *distance between charges*. The *product of the charges* is the same in both compounds because both sodium and potassium ions have a +1 charge. Attraction increases as distance decreases, so the ion with the smaller radius, Na^+, will form a stronger ionic interaction (**NaCl**).

2.78 Plan: Review the definitions of empirical and molecular formulas.
Solution:
An empirical formula describes the type and **simplest ratio** of the atoms of each element present in a compound whereas a molecular formula describes the type and **actual number** of atoms of each element in a molecule of the compound. The empirical formula and the molecular formula can be the same. For example, the compound formaldehyde has the molecular formula, CH_2O. The carbon, hydrogen, and oxygen atoms are present in the ratio of 1:2:1. The ratio of elements cannot be further reduced, so formaldehyde's empirical formula and molecular formula are the same. Acetic acid has the molecular formula, $C_2H_4O_2$. The carbon, hydrogen, and oxygen atoms are present in the ratio of 2:4:2, which can be reduced to 1:2:1. Therefore, acetic acid's empirical formula is CH_2O, which is different from its molecular formula. Note that the empirical formula does not *uniquely* identify a compound, because acetic acid and formaldehyde share the same empirical formula but are not the same compound.

2.84 Plan: The empirical formula describes the **simplest ratio** of the atoms of each element present in a compound. Examine the subscripts and see if there is a common divisor. If one exists, divide all subscripts by this value.
Solution:
a) To find the empirical formula for N_2H_4, divide the subscripts by the highest common divisor, 2: N_2H_4 becomes $\mathbf{NH_2}$.

b) To find the empirical formula for $C_6H_{12}O_6$, divide the subscripts by the highest common divisor, 6: $C_6H_{12}O_6$ becomes $\mathbf{CH_2O}$.

2.86 Plan: Locate each of the individual elements on the periodic table, and assign charges to each of the ions. For A group cations (metals), ion charge = group number; for anions (nonmetals), ion charge = group number minus 8. Find the smallest number of each ion that gives a neutral compound.
Solution:
a) Sodium is a metal that forms a +1 (Group **1**A) ion and nitrogen is a nonmetal that forms a –3 ion (Group **5**A, 5 – 8 = –3).

$$\overset{+1}{Na}\ \overset{-3}{N} \qquad \overset{+3\ -3}{\overset{+1}{Na_3}N}$$

The compound is $\mathbf{Na_3N}$**, sodium nitride**.

b) Oxygen is a nonmetal that forms a –2 ion (Group **6**A, 6 – 8 = –2) and strontium is a metal that forms a +2 ion (Group **2**A).

$$\overset{+2}{Sr}\ \overset{-2}{O}$$

The compound is **SrO, strontium oxide**.

c) Aluminum is a metal that forms a +3 ion (Group **3**A) and chlorine is a nonmetal that forms a –1 ion (Group **7**A, 7– 8 = –1).

$$\overset{+3}{Al}\ \overset{-1}{Cl} \qquad \overset{+3\ -3}{\overset{+3}{Al}\overset{-1}{Cl_3}}$$

The compound is $\mathbf{AlCl_3}$**, aluminum chloride**.

2.94 Plan: Review the rules for nomenclature covered in the chapter. Compounds must be neutral.
Solution:
a) Barium [Group **2**A(2)] forms Ba^{2+} and oxygen [Group 6A(16)] forms O^{2-} (6 – 8 = –2) so the neutral compound forms from one Ba^{2+} ion and one O^{2-} ion. Correct formula is **BaO**.
b) Iron(II) indicates Fe^{2+} and nitrate is NO_3^- so the neutral compound forms from one iron(II) ion and two nitrate ions. Correct formula is $\mathbf{Fe(NO_3)_2}$.
c) Mn is the symbol for manganese. Mg is the correct symbol for magnesium. Correct formula is **MgS**. Sulfide is the S^{2-} ion and sulfite is the SO_3^{2-} ion.

2.96 Plan: Acids donate H^+ ion to solution, so the acid is a combination of H^+ and a negatively charged ion.
Binary acids (H plus one other nonmetal) are named hydro + nonmetal root + ic acid
Oxoacids (H + an oxoanion) are named by changing the suffix of the oxoanion: -ate becomes -ic acid and -ite becomes -ous acid.
Solution:
a) Hydrogen sulfate is HSO_4^-, so its source acid is **H_2SO_4**. Name of acid is **sulfuric acid** (-ate becomes -ic acid).
b) **HIO_3, iodic acid** (IO_3^- is the iodate ion: -ate becomes -ic acid)
c) Cyanide is CN^-; its source acid is **HCN hydrocyanic acid** (binary acid).
d) **H_2S, hydrosulfuric acid** (binary acid)

2.100 Plan: This compound is composed of two nonmetals. Rule 1 ("Names and Formulas of Binary Covalent Compounds") indicates that the element with the lower group number is named first. Greek numerical prefixes are used to indicate the number of atoms of each element in the compound.
Solution:
Disulfur tetrafluoride **S_2F_4** di- indicates two S atoms and tetra- indicates four F atoms.

2.106 Plan: Review the rules of nomenclature and then assign a name. The molecular (formula) mass is the sum of the atomic masses of all of the atoms.
Solution:
a) **$(NH_4)_2SO_4$** ammonium is NH_4^+ and sulfate is SO_4^{2-}

N	=	2(14.01 amu)	=	28.02 amu
H	=	8(1.008 amu)	=	8.064 amu
S	=	1(32.07 amu)	=	32.07 amu
O	=	4(16.00 amu)	=	64.00 amu
				132.15 amu

b) **NaH_2PO_4** sodium is Na^+ and dihydrogen phosphate is $H_2PO_4^-$

Na	=	1(22.99 amu)	=	22.99 amu
H	=	2(1.008 amu)	=	2.016 amu
P	=	1(30.97 amu)	=	30.97 amu
O	=	4(16.00 amu)	=	64.00 amu
				119.98 amu

c) **$KHCO_3$** potassium is K^+ and bicarbonate is HCO_3^-

K	=	1(39.10 amu)	=	39.10 amu
H	=	1(1.008 amu)	=	1.008 amu
C	=	1(12.01 amu)	=	12.01 amu
O	=	3(16.00 amu)	=	48.00 amu
				100.12 amu

2.130 Plan: The rock is composed of 5.0% Fe_2SiO_4, 7.0% Mg_2SiO_4, and 88.0% SiO_2. It is comprised of four elements — iron, magnesium, silicon, and oxygen. At first glance, one would expect that silicon and oxygen would have the highest mass percentages because the rock is mostly silicon dioxide.

Solution:

1) First, determine the fraction of each element in each mineral.

Molecular mass of Fe_2SiO_4 = 2(55.85 amu) + 28.09 amu + 4(16.00 amu) = 203.79 amu

$$\text{Mass fraction of Fe} = \frac{\text{mass Fe}}{\text{molecular mass}} = \frac{2 \times 55.85 \text{ amu}}{203.79 \text{ amu}} = 0.5481$$

$$\text{Mass fraction of Si} = \frac{\text{mass Si}}{\text{molecular mass}} = \frac{1 \times 28.09 \text{ amu}}{203.79 \text{ amu}} = 0.1378$$

$$\text{Mass fraction of O} = \frac{\text{mass O}}{\text{molecular mass}} = \frac{4 \times 16.00 \text{ amu}}{203.79 \text{ amu}} = 0.3140$$

Molecular mass of Mg_2SiO_4 = 2(24.31 amu) + 28.09 amu + 4(16.00 amu) = 140.71 amu

$$\text{Mass fraction of Mg} = \frac{\text{mass Mg}}{\text{molecular mass}} = \frac{2 \times 24.31 \text{ amu}}{140.71 \text{ amu}} = 0.3455$$

$$\text{Mass fraction of Si} = \frac{\text{mass Si}}{\text{molecular mass}} = \frac{1 \times 28.09 \text{ amu}}{140.71 \text{ amu}} = 0.1996$$

$$\text{Mass fraction of O} = \frac{\text{mass O}}{\text{molecular mass}} = \frac{4 \times 16.00 \text{ amu}}{140.71 \text{ amu}} = 0.4548$$

Molecular mass of SiO_2 = 28.09 amu + 2(16.00 amu) = 60.09 amu

$$\text{Mass fraction of Si} = \frac{\text{mass Si}}{\text{molecular mass}} = \frac{1 \times 28.09 \text{ amu}}{60.09 \text{ amu}} = 0.4675$$

$$\text{Mass fraction of O} = \frac{\text{mass O}}{\text{molecular mass}} = \frac{2 \times 16.00 \text{ amu}}{60.09 \text{ amu}} = 0.5325$$

2) The percent mass of each element in the rock can be found by multiplying the percent of each element in each mineral by the percent of that mineral in the rock.

% Fe = (0.050) (0.5481) (100%) = **2.7% Fe**

% Mg = (0.070) (0.3455) (100%) = **2.4% Mg**

% Si = [(0.050) (0.1378) + (0.070) (0.1996) + (0.880) (0.4675)] (100%) = **43.2% Si**

% O = [(0.050) (0.3140) + (0.070) (0.4548) + (0.880) (0.5325)] (100%) = **51.6% O**

3) The percentages add up to ~100% (rounding accounts for the small error) and the majority of the rock is silicon and oxygen as expected.